SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code: ITA03	**Course Title:** Mobile computing
CO Assessed: CO1	**Assessment:** Assessment Tool 1 – Concept Map
Weightage: 60 % of CIA	**Total Marks:** 20 marks

CO1 Statement:

Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.

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Section / Batch: 2023	**Date:** 15/10/2026

Code of Conduct

I B.Lavanya / 192372046 7 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B.Lavanya

Annexure A – QUESTIONS AND ANSWERS

Question 1

Design a concept map that evaluates the significance of the fundamental concepts of mobile computing.

- Establish relationships among mobility, portability, wireless communication, and ubiquitous computing.
- Prioritize the concepts based on their contribution to mobile computing.
- Justify the linking phrases used between the concepts.

Answer – Concept Map

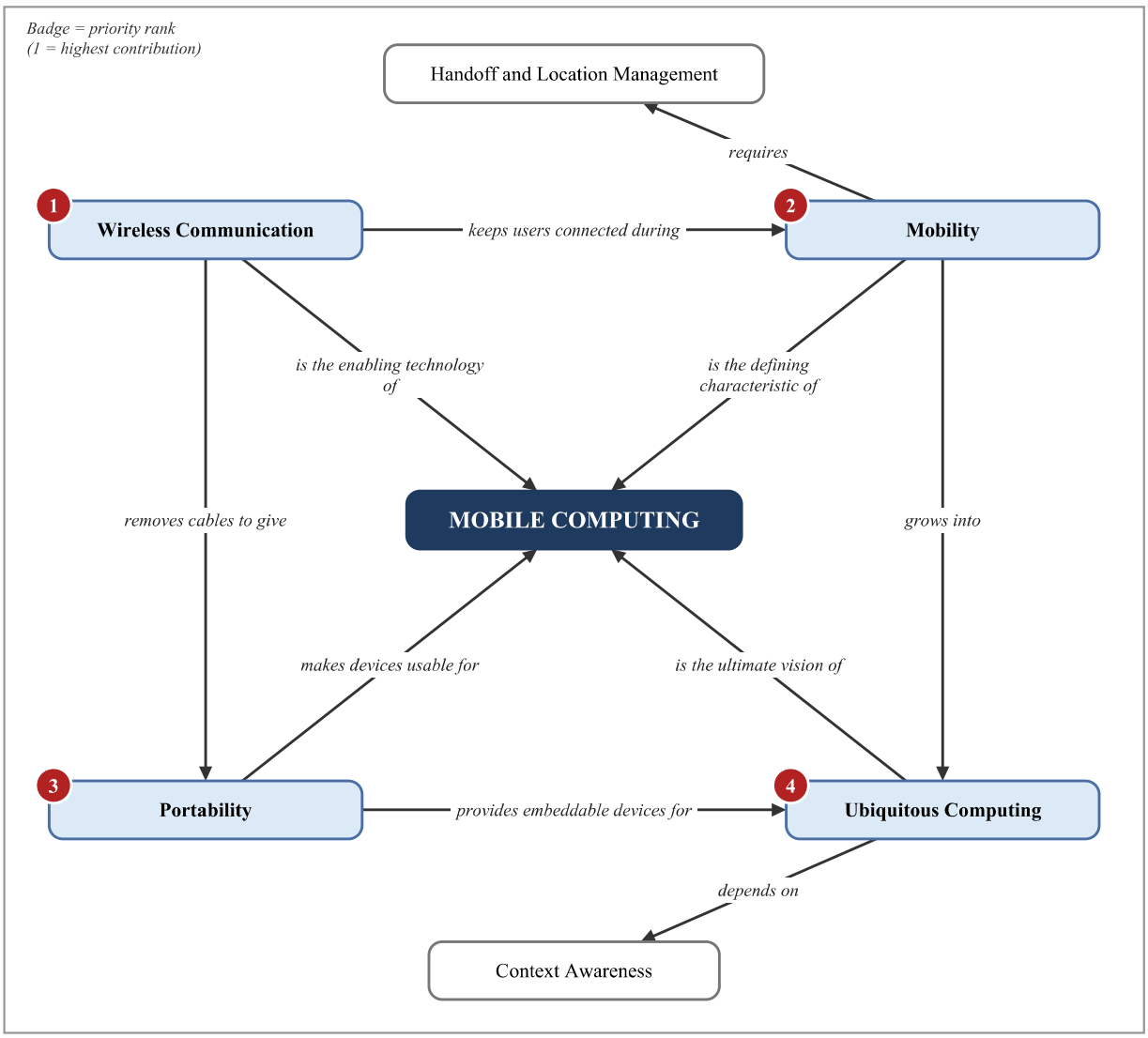


Figure 1: Concept map of the fundamental concepts of mobile computing

Priority of concepts (based on contribution to mobile computing)

Rank	Concept	Contribution and reason for priority
1	Wireless Communication	The foundation. Without a wireless medium (cellular, Wi-Fi, Bluetooth) no data can be exchanged while the user is on the move, so every other concept depends on it.

Rank	Concept	Contribution and reason for priority
2	Mobility	The defining feature of mobile computing. Users, devices and sessions change location yet stay connected, which drives handoff, roaming and location management.
3	Portability	A device-level property (small size, low weight, battery power). It makes the equipment practical to carry, but a portable device with no wireless link is only a stand-alone gadget.
4	Ubiquitous Computing	The long-term vision: computing available anywhere, anytime and invisibly. It is built on the other three concepts, so it is an outcome rather than a prerequisite.

Justification of linking phrases

Linking phrase (from - to)	Justification
Wireless Communication – "is the enabling technology of" – Mobile Computing	"Enabling" shows a necessary precondition: mobile computing cannot exist without a wireless link.
Mobility – "is the defining characteristic of" – Mobile Computing	Mobility is what separates mobile computing from fixed-network computing.
Portability – "makes devices usable for" – Mobile Computing	Compact, battery-powered terminals are the means through which mobile computing is carried out.
Ubiquitous Computing – "is the ultimate vision of" – Mobile Computing	Shows direction: mobile computing evolves towards pervasive, invisible computing.
Wireless Communication – "keeps users connected during" – Mobility	Continuous wireless connectivity lets a session survive while the user moves.
Mobility – "grows into" – Ubiquitous Computing	Access from any location (mobility) extends naturally to access everywhere and at all times.
Portability – "provides embeddable devices for" – Ubiquitous Computing	Miniaturised devices can be embedded in everyday objects, which ubiquitous computing requires.
Wireless Communication – "removes cables to give" – Portability	Removing wired connections is what lets a device be carried around freely.
Mobility – "requires" – Handoff and Location Management	A moving user changes cells, so the network must hand over calls and track the user.
Ubiquitous Computing – "depends on" – Context Awareness	Pervasive systems must sense location, time and user situation to adapt their behaviour.

Conclusion: Wireless communication contributes most because it enables the rest, followed by mobility (the defining property), portability (the device enabler) and ubiquitous computing (the end goal).

Question 2

Construct a concept map that evaluates the effectiveness of the cellular network architecture.

- Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem.
- Identify the most critical components influencing communication.
- Justify your evaluation through meaningful concept relationships.

Answer – Concept Map

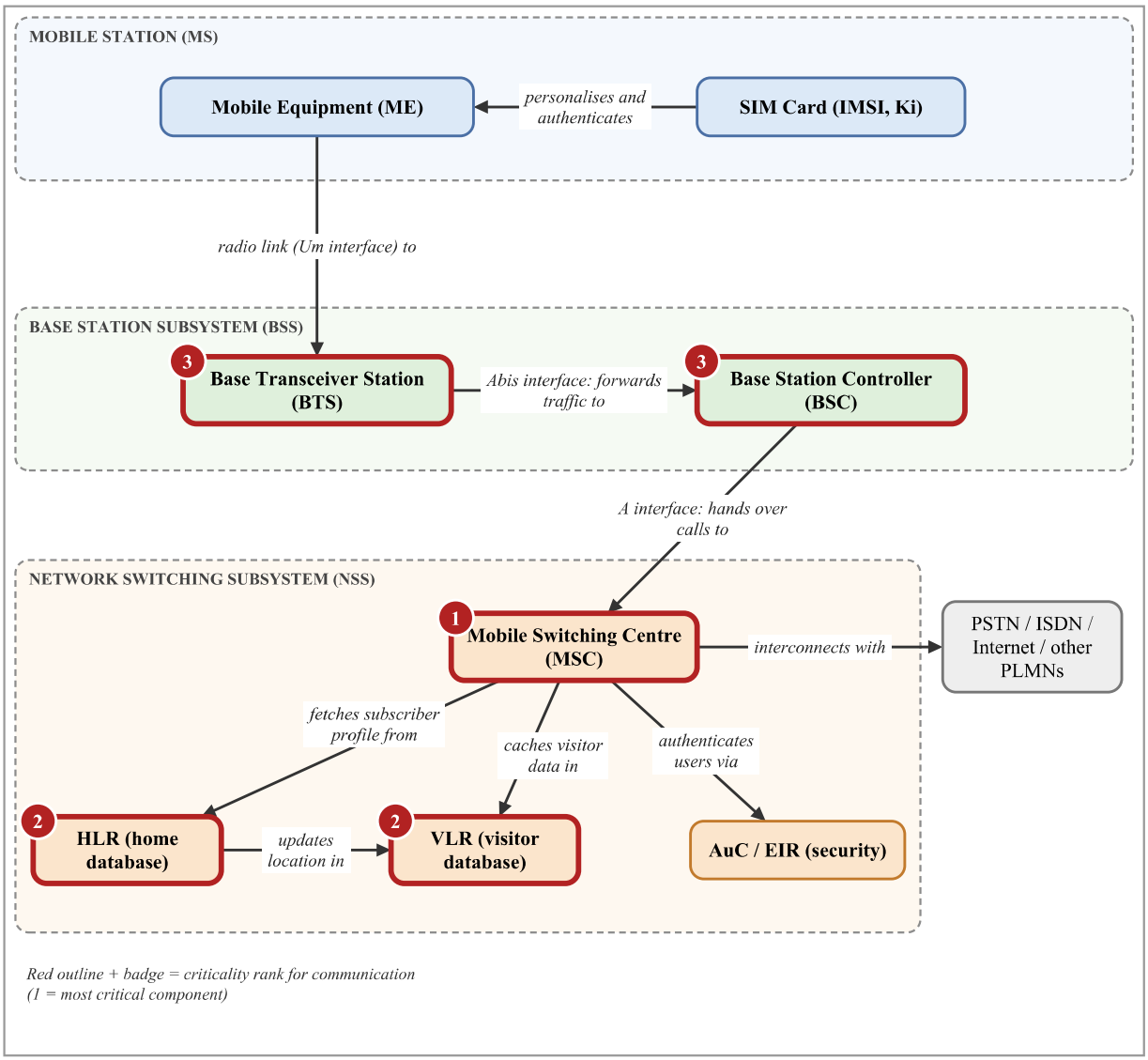


Figure 2: Concept map of the cellular (GSM) network architecture

Most critical components influencing communication

Rank	Component	Why it is critical
1	MSC	Core switching element: sets up, routes and releases calls, performs handover between BSCs and links to external networks. Its failure stops service for the whole area.
2	HLR / VLR	Hold the permanent and temporary subscriber and location data. Without them calls cannot be routed to the right cell, so mobility management fails.

Rank	Component	Why it is critical
3	BSC / BTS	Provide the radio link and radio-resource control. A BTS failure affects one cell and a BSC failure the cells under it, a smaller impact than the NSS.

Justification of relationships

Linking phrase (from - to)	Justification
SIM Card – "personalises and authenticates" – Mobile Equipment	The SIM stores the IMSI and key Ki, tying the handset to one subscriber.
Mobile Equipment – "radio link (Um interface) to" – BTS	The air interface is the only wireless hop between the user and the network.
BTS – "Abis interface: forwards traffic to" – BSC	The BTS handles radio transmission; the BSC controls many BTSs and manages channels and handover.
BSC – "A interface: hands over calls to" – MSC	Connects the radio part (BSS) to the switching part (NSS).
MSC – "fetches subscriber profile from" – HLR	The HLR holds the permanent profile and current location of every subscriber.
MSC – "caches visitor data in" – VLR	The VLR keeps temporary data of subscribers roaming in the MSC area, avoiding repeated HLR queries.
HLR – "updates location in" – VLR	Location updates keep both databases consistent so calls reach the user.
MSC – "authenticates users via" – AuC / EIR	Checks the subscriber identity and equipment validity, ensuring secure access.
MSC – "interconnects with" – PSTN / other PLMNs	Gateway functions allow calls to and from fixed and other mobile networks.

Conclusion: The architecture is effective because each subsystem has a clear role (radio access, radio control, switching and mobility management) and standard interfaces (Um, Abis, A). The NSS, and especially the MSC with the HLR/VLR, is the most critical part, as every call and every location update passes through it.

Question 3

Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.

- Illustrate their interdependencies.
- Assess which subsystem has the greatest influence on service continuity.
- Justify your conclusions using concept relationships.

Answer – Concept Map

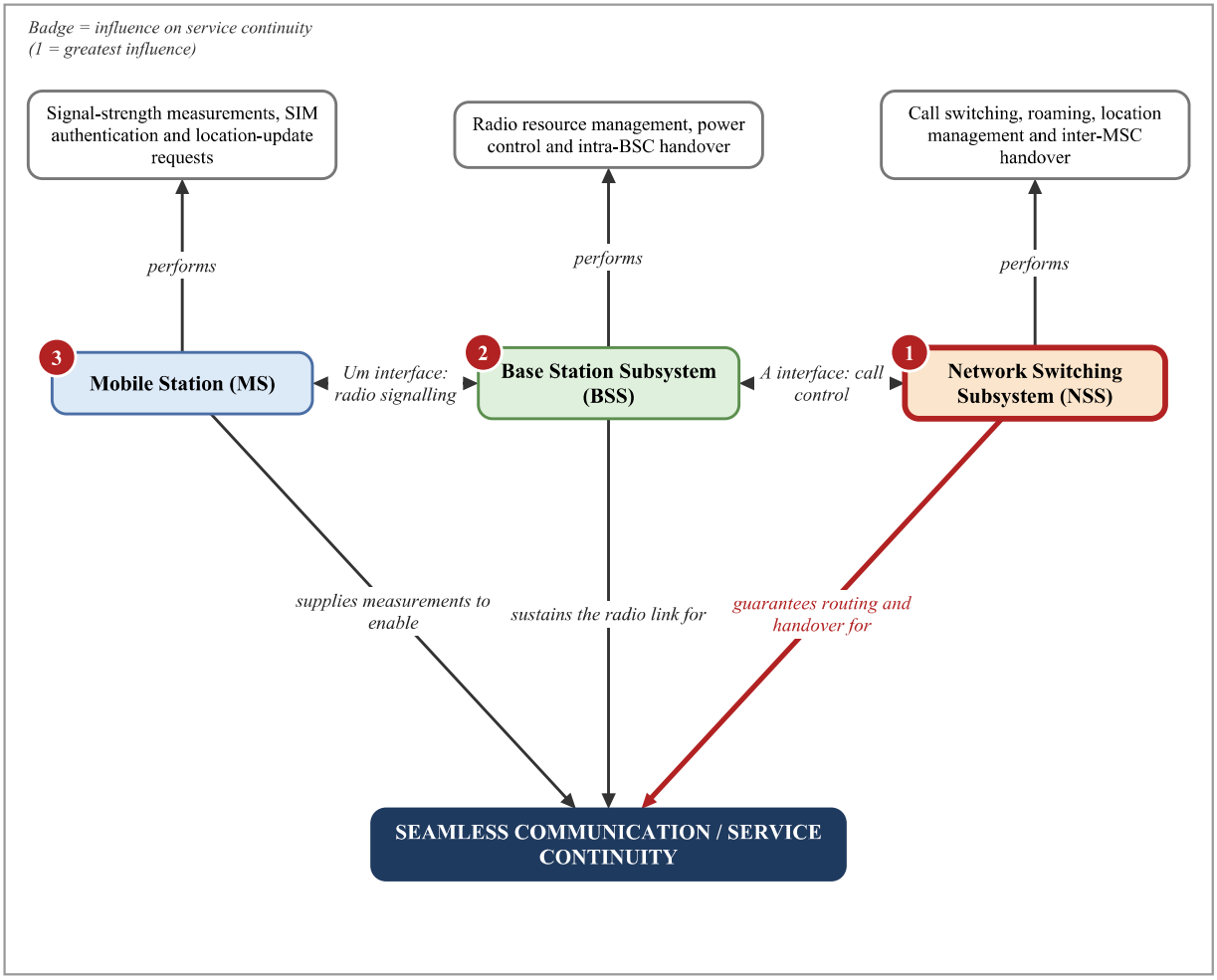


Figure 3: Contribution and interdependency of MS, BSS and NSS in seamless communication

Contribution and interdependency

Subsystem	Contribution to seamless communication	Depends on
MS	Measures signal strength of the serving and neighbouring cells, authenticates through the SIM and sends location-update requests.	BSS for radio access; NSS for authentication and routing.
BSS	Allocates radio channels, controls transmit power and performs intra-BSC handover to keep the radio link alive.	MS measurement reports; NSS for call control and inter-BSC handover.
NSS	Switches calls, manages location through HLR/VLR, supports roaming and executes inter-MSC handover.	BSS to reach the user; MS location updates to know where to route calls.

Subsystem with the greatest influence on service continuity

The NSS has the greatest influence (rank 1), followed by the BSS (rank 2) and the MS (rank 3). The BSS can handle a handover inside its own area, but handover between BSCs or MSCs, roaming, location updates and re-routing of calls are performed only by the NSS (MSC, VLR, HLR). A failed MS affects a single user and a failed BTS a single cell, whereas a failure in the NSS interrupts service for the whole region. This is why the link from the NSS to Seamless Communication is drawn in red.

Justification of linking phrases

Linking phrase	Justification
MS / BSS / NSS – "performs" – function box	Connects each subsystem to the specific task that it contributes.
MS and BSS (two-way) – "Um interface: radio signalling"	Two-way arrow: the MS reports measurements and the BSS returns channel and handover commands.
BSS and NSS (two-way) – "A interface: call control"	Two-way arrow: the BSS passes requests up and the NSS returns routing and handover decisions.
MS – "supplies measurements to enable" – Seamless Communication	Handover decisions cannot be made without the MS measurement reports.
BSS – "sustains the radio link for" – Seamless Communication	Radio resource management keeps the connection within and between cells.
NSS – "guarantees routing and handover for" – Seamless Communication	The NSS keeps the call continuous across cells, MSCs and networks, so its link has the highest weight.

Question 4

Develop a concept map that evaluates the effectiveness of the mobile network protocol stack.

- Illustrate interactions among protocol layers.
- Assess how each layer contributes to reliable mobile communication.
- Support your evaluation with meaningful hierarchical relationships.

Answer – Concept Map

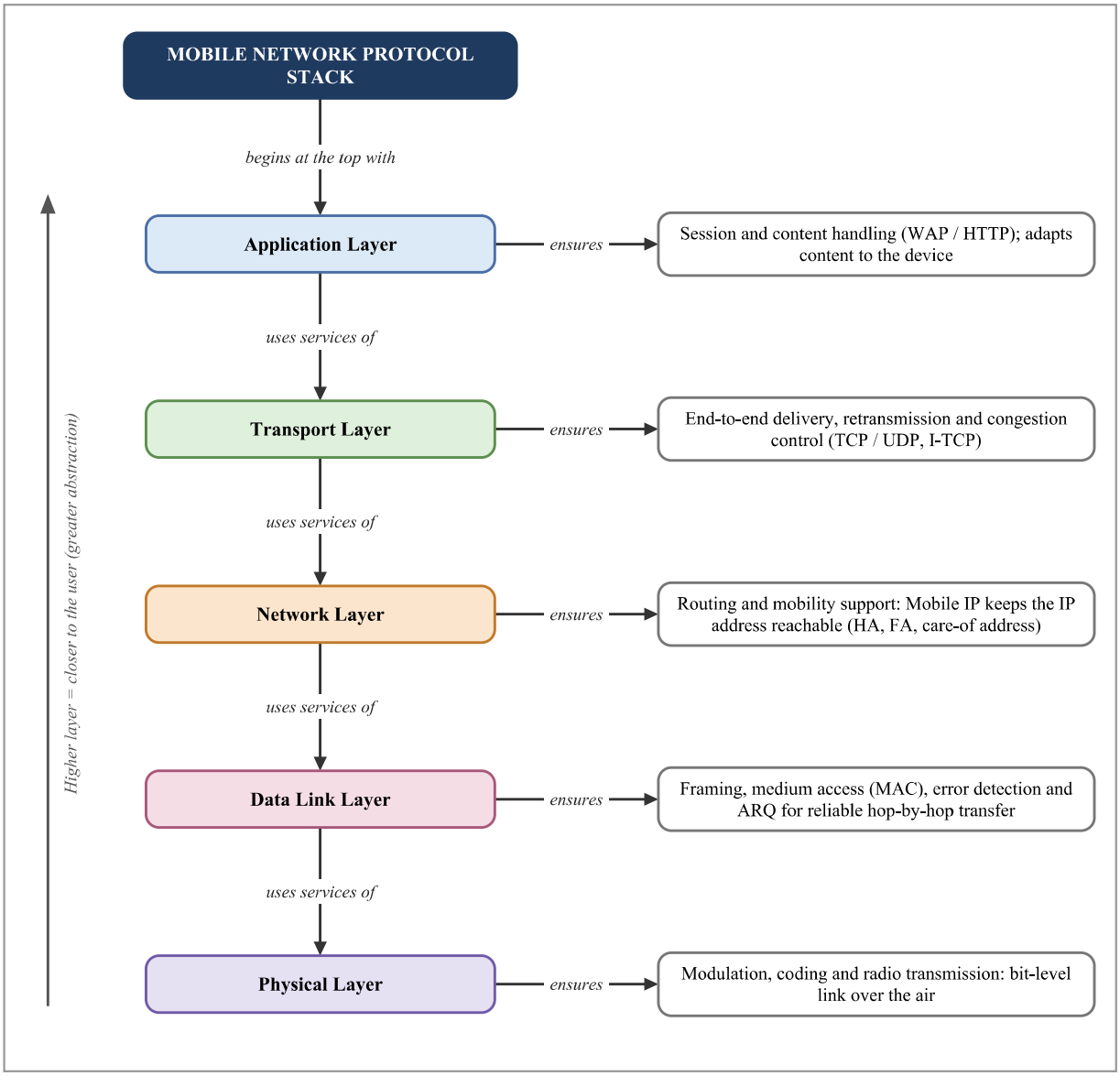


Figure 4: Hierarchical concept map of the mobile network protocol stack

Contribution of each layer to reliable mobile communication

Layer	Key protocols / functions	Contribution to reliability
Application	WAP, HTTP, WSP	Manages sessions and adapts content to small screens and low bandwidth, so the user gets usable and complete content.
Transport	TCP, UDP, Indirect TCP, Snooping TCP	Provides end-to-end delivery with acknowledgements, retransmission and congestion control. Mobile variants stop wireless errors being mistaken for congestion.

Layer	Key protocols / functions	Contribution to reliability
Network	IP, Mobile IP (HA, FA, care-of address), routing	Keeps the node reachable at its home address while it moves between networks, so sessions are not broken.
Data Link	MAC (TDMA / CDMA / CSMA-CA), LLC, ARQ	Controls access to the shared radio medium, frames the data and detects and corrects errors on each hop.
Physical	Modulation, coding, FDMA / TDMA radio	Transmits bits over the air and handles noise and fading using coding and modulation.

Hierarchical relationships

The stack is hierarchical: each layer uses the services of the layer directly below it ("uses services of") and offers its own service to the layer above. Data moves down through encapsulation on the sender and up through decapsulation on the receiver. Reliability is therefore built in stages: the physical layer provides the raw link, the data link layer makes each hop dependable, the network layer keeps the node reachable while moving, the transport layer guarantees end-to-end delivery and the application layer adapts content for the user. A weakness in a lower layer (for example fading) is corrected by the layers above, which makes the design effective for mobile communication.

Question 5

Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).

- Relate operators, subscribers, switching centers, and radio access networks.
- Assess how these components collectively provide seamless connectivity.
- Justify the relationships included in the concept map.

Answer – Concept Map

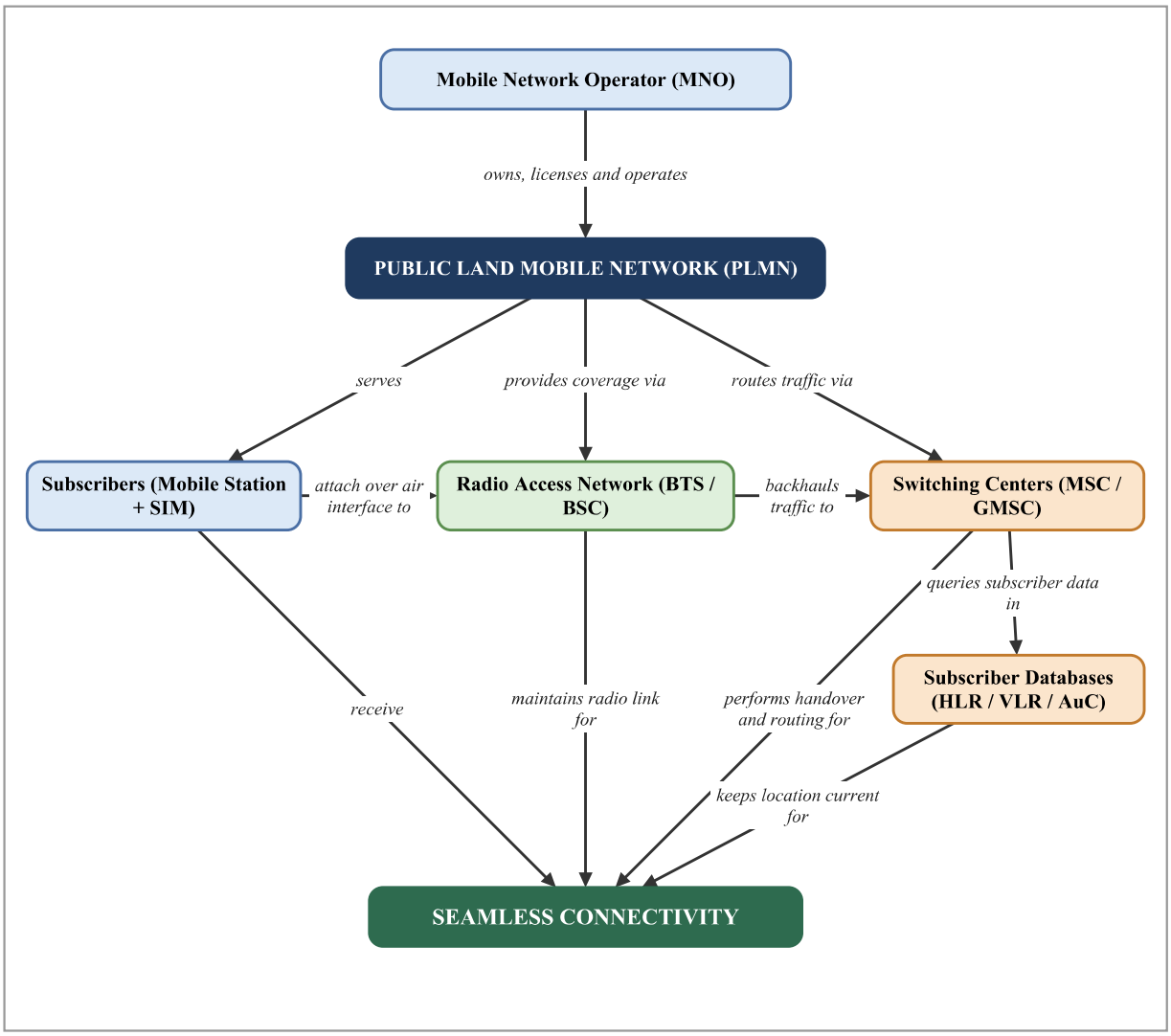


Figure 5: Concept map of a Public Land Mobile Network (PLMN)

Collective contribution to seamless connectivity

Component	Role	Contribution to seamless connectivity
Operator (MNO)	Owns the licence and spectrum and operates the PLMN.	Plans coverage and capacity and manages roaming agreements, so subscribers get service across areas.
Subscribers (MS + SIM)	Users of the network, identified by IMSI.	Send measurement reports and location updates that let the network track and serve them.

Component	Role	Contribution to seamless connectivity
Radio Access Network	BTS and BSC provide radio coverage and manage radio resources.	Maintains the radio link and handles handover between cells.
Switching Centers	MSC and GMSC switch calls and link to other networks.	Route calls to the current location of the subscriber and perform handover between MSCs.
Subscriber Databases	HLR, VLR and AuC store profiles, location and security keys.	Keep location and authentication data current so calls always reach the right place.

Justification of relationships

Linking phrase	Justification
Operator – "owns, licenses and operates" – PLMN	A PLMN is defined by the operator that holds the licence and runs the network.
PLMN – "serves" – Subscribers	The purpose of the network is to provide service to its subscribers.
PLMN – "provides coverage via" – RAN; "routes traffic via" – Switching Centers	Shows that the PLMN is made up of the radio access part and the switching part.
Subscribers – "attach over air interface to" – RAN	The RAN is the point where users connect to the network wirelessly.
RAN – "backhauls traffic to" – Switching Centers	Radio traffic is carried to the MSC for switching and call control.
Switching Centers – "queries subscriber data in" – Databases	Routing and authentication depend on HLR/VLR data.
RAN / Switching Centers / Databases – "maintains", "performs", "keeps" – Seamless Connectivity	Each component supports continuity in a different way; together they deliver an uninterrupted service.

Conclusion: The PLMN is operationally efficient because the RAN, switching centers and databases work together under the operator: the RAN keeps the radio link, the switching centers route calls and perform handover, and the databases keep location data current, so subscribers stay connected while moving.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____